

BELLABEAT CASE STUDY REPORT

EXECUTIVE SUMMARY

This analysis examined user activity, sleep behaviour, calorie expenditure, heart rate observations, and weight records from Fitbit smart device users.

The objective was to identify behavioural patterns that could support data-driven recommendations for Bellabeat's wellness products and marketing strategies.

Following data cleaning, feature engineering, relational integration, and exploratory analysis, several meaningful trends emerged regarding physical activity, sleep habits, and user engagement.

The analysis identified substantial variation in physical activity and sleep behaviour across users. A strong positive relationship was observed between daily step counts and calorie expenditure, while activity segmentation revealed that many users remain within sedentary or moderately active categories. These findings suggest opportunities for personalized coaching, behavioural segmentation, and enhanced wellness engagement strategies.

PROJECT INTRODUCTION

Bellabeat is a high-tech wellness company that develops smart products designed to help women monitor and improve their health. As the company seeks to expand its market presence, understanding how consumers interact with wearable fitness devices can provide valuable insights for product development and marketing strategy.

This project analyses Fitbit fitness tracker data to explore user behaviour related to physical activity, sleep patterns, calorie expenditure, heart rate monitoring, and weight tracking. Through data preparation, feature engineering, exploratory analysis, and business-focused interpretation, the study aims to identify behavioural trends that can support data-driven decision-making.

The analysis follows a structured analytics workflow, including data cleaning, validation, relational integration, exploratory data analysis, and strategic recommendation development. The findings are intended to demonstrate how wearable device data can be transformed into actionable business insights that support customer engagement and wellness-focused product innovation.

BUSINESS TASK

Analyse smart device usage data to identify behavioural trends and generate actionable recommendations that can inform Bellabeat's marketing strategy and product development initiatives.

PROJECT SCOPE

This project focuses on descriptive analytics and behavioural exploration of Fitbit smart-device data. The objective is to understand user activity patterns, sleep behaviour, and wellness trends rather than develop predictive or machine learning models.

DATASET OVERVIEW

The analysis utilizes the Fitbit Fitness Tracker Dataset, which contains anonymized smart-device usage records collected from Fitbit users. The Fitbit Fitness Tracker Dataset was obtained from Kaggle and is commonly used in analytics case studies to examine behavioural patterns associated with wearable health technology. The dataset captures multiple dimensions of user wellness behaviour, including physical activity, sleep patterns, calorie expenditure, heart rate measurements, and weight tracking.

The analytical environment consists of six datasets:

Dataset	Description	Granularity
Daily Activity	Daily activity and calorie metrics	User-Day
Sleep	Sleep duration and time in bed	User-Day
Hourly Steps	Hourly step counts	User-Hour
Hourly Calories	Hourly calorie expenditure	User-Hour
Heart Rate	Heart rate measurements	User-Second
Weight Log	Weight and BMI tracking events	Event-Based

Collectively, these datasets provide a comprehensive view of user health and activity behaviours across multiple temporal resolutions, enabling the analysis of activity patterns, sleep habits, physiological indicators, and wellness trends.

TOOLS & TECHNOLOGIES

The analysis was conducted using the following tools and technologies:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Jupyter Notebook
- Visual Studio Code

These tools were used for data preparation, feature engineering, relational integration, exploratory analysis, data visualization, and reporting.

METHODOLOGY

1: Structural Profiling Summary

A comprehensive structural profiling assessment was conducted across all six Fitbit datasets to evaluate data integrity, schema consistency, completeness, and analytical readiness prior to preprocessing and exploratory analysis. The profiling stage focused on understanding dataset granularity, variable structure, datatype validity, missing value distribution, duplication risk, and computational characteristics.

The analytical environment consists of six relational datasets representing different dimensions of user wellness behaviour:

1. Daily_activity
2. Sleep
3. Hourly_steps
4. Hourly_calories
5. Heart_rate
6. Weight_log

These datasets collectively capture user activity, sleep behaviour, calorie expenditure, heart rate monitoring, and weight tracking across multiple temporal resolutions, including daily, hourly, and second-level granularity.

Structural inspection revealed that the datasets are generally well-organized and analytically coherent. Most variables were successfully interpreted using appropriate numerical datatypes (`'int64'` and `'float64'`), particularly for activity metrics, physiological measurements, and distance-related variables. However, temporal attributes such as `'ActivityDate'`, `'SleepDay'`, `'ActivityHour'`, and `'Time'` were initially stored as string (`'object'`) datatypes rather than native datetime formats. This creates limitations for temporal aggregation, time-series analysis, and feature engineering, necessitating datetime normalization during preprocessing.

Data quality assessment showed relatively strong structural integrity across the datasets. Most tables exhibited low or negligible missing-value distributions, particularly within the core behavioural datasets (`'daily_activity'`, `'hourly_steps'`, and `'hourly_calories'`). Duplicate auditing similarly indicated minimal row-level duplication, suggesting that the exported Fitbit records maintain acceptable transactional consistency.

The datasets also exhibit varying levels of granularity and analytical complexity. Daily-level datasets (`'Daily_activity'` and `'sleep'`) provide aggregated behavioural summaries suitable for high-level wellness analysis and user segmentation. Hourly datasets (`'hourly_steps'` and `'hourly_calories'`) introduce finer temporal resolution for activity-pattern analysis and intraday

behavioural profiling. In contrast, the `heart\_rate` dataset operates at second-level granularity, representing a high-frequency physiological time-series structure requiring aggregation and optimization prior to integration with lower-granularity tables. The `weight\_log` dataset functions as an event-based tracking table with comparatively sparse observations.

Memory profiling indicated that most datasets are computationally lightweight and suitable for in-memory processing. However, the second-level heart-rate dataset demonstrates substantially higher memory consumption relative to the remaining tables, highlighting the need for careful aggregation and datatype optimization during preprocessing.

Overall, the structural profiling stage confirms that the Fitbit data ecosystem possesses strong analytical potential with relatively high structural consistency and low integrity risk. Nevertheless, preprocessing operations remain necessary to standardize temporal datatypes, normalize schema conventions, validate numerical integrity, and align dataset granularity prior to relational merging and downstream exploratory analysis.

2: Data Cleaning Readiness Summary

The data quality validation process indicates that the Fitbit datasets exhibit strong structural integrity and are suitable for downstream analytical processing.

Datetime variables were successfully converted to native datetime formats, enabling temporal analysis, feature engineering, and time-based aggregation. Column names were standardized to improve consistency, readability, and maintainability across the analytical workflow.

Quality validation checks identified no material issues relating to negative activity measurements, implausible physiological observations, or invalid duration-based metrics. Duplicate assessment revealed three duplicate records within the sleep dataset, which were removed during preprocessing to preserve data integrity.

Missing value analysis identified substantial missingness within the `fat` variable of the weight log dataset (65 missing values out of 67 observations, approximately 97% missingness). Given the limited analytical value and high degree of incompleteness, the variable was excluded from further analysis, while all remaining weight-related attributes were retained.

Following duplicate management and missing-value treatment, all analytical datasets achieved complete records with no remaining missing values or duplicate observations. The datasets are therefore considered analytically ready for feature engineering, temporal enrichment, relational integration, and exploratory analysis.

Subsequent preprocessing efforts focus on deriving business-relevant features and aligning dataset granularity to support robust analytical modelling and insight generation.

3: Data Integration Strategy

Following the granularity assessment, datasets sharing common temporal resolutions were integrated to create analytical datasets.

The `daily_activity` and `sleep` datasets were merged using user identifier and activity date, resulting in a daily analytical dataset suitable for behavioural and sleep-related analysis.

Similarly, `hourly_steps` and `hourly_calories` were merged using user identifier and activity timestamp to support hourly behavioural analysis.

The `heart_rate` dataset was retained separately due to its second-level granularity, while the `weight_log` dataset was analysed independently because of its event-based structure and limited observation count.

This integration strategy preserved analytical integrity while minimizing duplication and row inflation.

4: Feature Engineering Summary

Several derived variables were created to enhance analytical capability.

These included:

- Day of Week
- Month
- Weekend Indicator
- Total Active Minutes
- Activity Level Segmentation
- Sleep Efficiency

These features enabled temporal analysis, behavioural segmentation, and wellness-focused insight generation throughout the exploratory analysis phase.

EDA SUMMARY

1: Users show a wide variation in daily step counts, indicating mixed activity lifestyles ranging from sedentary to highly active.

2: There is a positive relationship between step count and calories burned, confirming expected physiological behaviour patterns.

3: Sleep duration varies significantly across users, with several users not meeting recommended sleep thresholds.

4: No strong linear relationship is observed between physical activity and sleep duration, suggesting sleep quality may be influenced by additional factors.

5: Activity levels differ across weekdays and weekends, indicating behavioural shifts based on routine structure.

6: User segmentation reveals that a large proportion of users fall into moderate or sedentary activity categories.

Overall, the dataset provides strong evidence of behavioural diversity, making it suitable for targeted wellness recommendations.

MISSING VALUES

Sleep-related variables contained 530 missing observations after integration with the daily activity dataset. These missing values reflect the absence of corresponding sleep records rather than incomplete measurements. Consequently, no imputation was performed, and sleep analyses were conducted using available sleep observations only.

GRANULARITY ASSESSMENT SUMMARY

The Fitbit data ecosystem contains multiple temporal resolutions, ranging from daily aggregated activity records to second-level physiological observations.

The daily_activity and sleep datasets operate at a user-day granularity and are therefore suitable for direct integration. The hourly_steps and hourly_calories datasets share a common user-hour granularity, making them appropriate for hourly behavioural analysis.

The heart_rate dataset operates at a much finer user-second granularity and requires aggregation prior to relational integration to avoid row inflation and duplication. The weight_log dataset represents event-based observations and should be analysed independently unless a specific use case requires integration.

Based on the granularity assessment, daily-level and hourly-level datasets will be merged separately to preserve analytical integrity.

KEY FINDINGS

1: Significant Variation in Physical Activity

Daily step counts varied substantially across users, indicating the presence of distinct activity segments ranging from sedentary to highly active individuals.

2: Strong Relationship Between Steps and Calories Burned

Users with higher step counts consistently demonstrated higher calorie expenditure, reinforcing the effectiveness of movement-based activity tracking.

3: Sleep Patterns Vary Across Users

Sleep duration and sleep efficiency varied considerably among participants, suggesting opportunities for personalized sleep coaching and monitoring.

4: Behavioural Differences Across Days of the Week

Activity levels showed noticeable variation across weekdays and weekends, indicating that user routines influence wellness behaviours.

5: Majority of Users Are Not Consistently Highly Active

Activity segmentation revealed that many users fall within sedentary or moderately active categories, highlighting potential opportunities for behaviour-change interventions.

BUSINESS RECOMMENDATIONS

1: Promote Activity-Based Goal Setting

Bellabeat should emphasize personalized step goals and daily movement targets within its mobile application. Users exhibiting lower activity levels may benefit from adaptive coaching and motivational reminders.

2: Expand Sleep Coaching Features

Given the variability observed in sleep behaviour, Bellabeat can strengthen its value proposition through enhanced sleep analytics, bedtime recommendations, and recovery tracking.

3: Introduce Behavioural Segmentation

Users should be segmented according to activity levels (Sedentary, Moderately Active, Highly Active) to support personalized content, notifications, and wellness recommendations.

4: Leverage Weekly Behavioural Trends

Marketing campaigns and in-app challenges can be strategically scheduled around periods of lower engagement to encourage consistent wellness habits.

5: Develop Integrated Wellness Dashboards

Combining activity, sleep, and physiological metrics into a unified wellness score may help users better understand the relationship between daily behaviours and overall health outcomes.

PROJECT CONCLUSION

The Fitbit dataset provides valuable insight into user wellness behaviours, particularly in relation to physical activity, sleep habits, and calorie expenditure.

The analysis revealed substantial variation in activity levels and sleep patterns, highlighting opportunities for personalized health interventions.

By leveraging these behavioural insights, Bellabeat can enhance user engagement through targeted coaching, personalized wellness recommendations, and data-driven product development strategies.

Future analysis could incorporate additional physiological indicators such as aggregated heart rate metrics and longitudinal weight tracking to further strengthen wellness modelling capabilities.

LIMITATIONS

- 1: The dataset represents a limited sample of Fitbit users and may not fully represent the broader wearable device market.
- 2: The weight log dataset contained limited observations, restricting the depth of weight-related analysis.
- 3: Heart rate data was available at a much finer temporal granularity and was not fully integrated into the daily analytical layer.
- 4: The analysis focused primarily on descriptive analytics and does not establish causal relationships between variables.